```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud

df = pd.read_csv('C:\\Users\\njain121\\Desktop\\Scalar\\CaseStudy\\
netflix.csv',index_col = 0)
```

##Replace NaN values and split Cast column

```
df['cast'] = df['cast'].fillna('Unknown cast')
df['cast'] = df['cast'].apply(lambda x: x.split(','))
df = df.explode('cast').reset_index(drop=True)
df['cast'] = df['cast'].str.strip()
```

##Replace NaN values and split Listed_in(genre) column

```
df['listed_in'] = df['listed_in'].fillna('Unknown genre')
df['listed_in'] = df['listed_in'].apply(lambda x: x.split(','))
df = df.explode('listed_in').reset_index(drop=True)
df['listed_in'] = df['listed_in'].str.strip()
df.shape
```

```
(149512, 11)
```

```
df['country'] = df['country'].fillna('Unknown country')
df['country'] = df['country'].apply(lambda x: x.split(','))
df = df.explode('country').reset_index(drop=True)
df['country'] = df['country'].str.strip()
```

```
df['director'] = df['director'].fillna('Unknown director')
df['director'] = df['director'].apply(lambda x: x.split(','))
df = df.explode('director').reset_index(drop=True)
df['director'] = df['director'].str.strip()
df.shape
```

```
(202065, 11)
```

```
df['date_added'] = pd.to_datetime(df['date_added'], format='mixed')
df['week_number'] = df['date_added'].dt.isocalendar().week
df.head(2)
```

```
      type                 title          director          cast  \
0    Movie  Dick Johnson Is Dead   Kirsten Johnson  Unknown cast
1  TV Show         Blood & Water  Unknown director    Ama Qamata

          country date_added  release_year rating   duration  \
0  United States 2021-09-25          2020  PG-13     90 min
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons
```

```
                 listed_in
description  \
0           Documentaries  As her father nears the end of his life,
filmm...
1  International TV Shows  After crossing paths at a party, a Cape
Town t...

   week_number
0           38
1           38
```

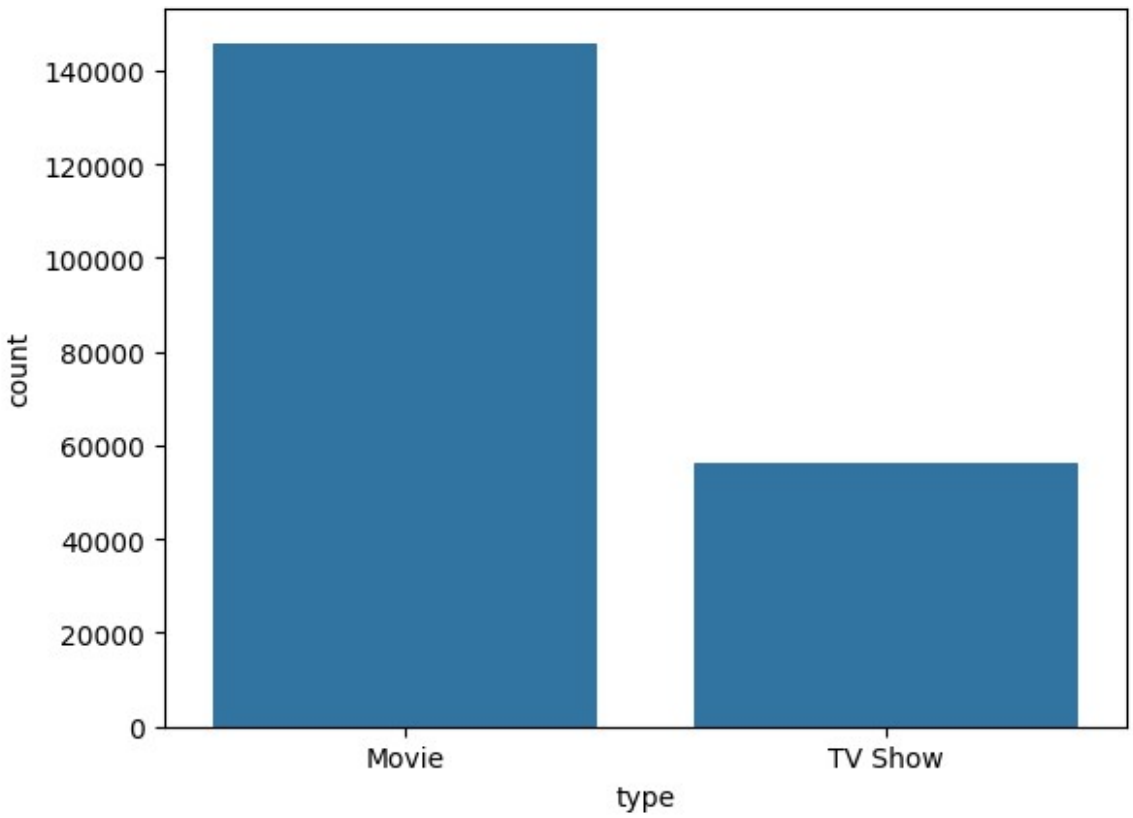
```
df['month'] = df['date_added'].dt.month
df['month'] = df['month'].fillna('0')
df['month'] = df['month'].astype(int)
df.head(2)
```

```
      type                  title          director          cast  \
0    Movie  Dick Johnson Is Dead   Kirsten Johnson  Unknown cast
1  TV Show         Blood & Water  Unknown director    Ama Qamata

         country date_added  release_year rating   duration  \
0  United States 2021-09-25          2020  PG-13     90 min
1   South Africa 2021-09-24          2021  TV-MA  2 Seasons

                 listed_in
description  \
0           Documentaries  As her father nears the end of his life,
filmm...
1  International TV Shows  After crossing paths at a party, a Cape
Town t...

   week_number  month
0           38      9
1           38      9
```

```
type_count = df['type'].value_counts()
type_count
```

```
type
Movie      145917
TV Show     56148
Name: count, dtype: int64
```

```
cp = sns.countplot(data=df, x='type', order = type_count.index)
plt.show()
```



Observations 1:

1. We have more number of movies on netflix than TV shows

```
df_movies = df[df['type'] == 'Movie']

top10_countries_Movies = df_movies['country'].value_counts().head(10)
top10_countries_Movies

country
United States      45817
India              21411
United Kingdom      8580
France              6607
Unknown country     6199
Canada              5738
Japan               3525
Spain               3469
Germany             3427
China               2377
Name: count, dtype: int64

df_TV = df[df['type'] == 'TV Show']
#df_TV.head()
```

```
top10_countries_TV = df_TV['country'].value_counts().head(10)
top10_countries_TV
```

```
country
United States      13533
Unknown country     5698
Japan               5154
United Kingdom      4385
South Korea         3754
Canada              2177
Mexico              2018
Spain               1846
Taiwan              1719
France              1647
Name: count, dtype: int64
```

Observations 2:

1. US and India are top 10 countries who watches movies on netflix
2. UK watches mostly TV shows on netflix

```
count_per_week_Movies = df_movies['week_number'].value_counts()
count_per_month_Movies = df_movies['month'].value_counts()
count_per_week.head(1),  count_per_month_Movies.head(1)
```

```
(week_number
 1    7838
 Name: count, dtype: Int64,
 month
 7    15075
 Name: count, dtype: int64)
```

Observations 3:

1. Week 1 or 7th month of the year is best to launch any movie on Netflix

```
count_per_week_TV = df_TV['week_number'].value_counts()
count_per_month_TV = df_TV['month'].value_counts()
count_per_week_TV.head(1), count_per_month_TV.head(1)
```

```
(week_number
 27    1977
 Name: count, dtype: Int64,
 month
 12    5498
 Name: count, dtype: int64)
```

Observations 4:

1. Week 27 or 12th month of the year is best to launch any TV show on Netflix

```
df_movies_fil = df_movies[df_movies['director'] != 'Unknown director']
count_per_director_Movies = df_movies_fil['director'].value_counts()
count_per_director_Movies.head(10)
```

```
director
Martin Scorsese        419
Youssef Chahine        409
Cathy Garcia-Molina    356
Steven Spielberg       355
Lars von Trier         336
Raja Gosnell           308
Tom Hooper             306
McG                    293
David Dhawan           270
Wilson Yip             260
Name: count, dtype: int64
```

Observations 5:

1. Director Martin Scorsese has directed maximum no. of movies which are present on Netflix

```
df_TV_fil = df_TV[df_TV['director'] != 'Unknown director']
count_per_director_TV = df_TV_fil['director'].value_counts()
count_per_director_TV.head(10)
```

```
director
Noam Murro           189
Thomas Astruc        160
Alan Poul            104
Laïla Marrakchi      104
Houda Benyamina      104
Damien Chazelle      104
Rob Seidenglanz      103
Alejandro Lozano      90
Jay Oliva             81
Manolo Caro           78
Name: count, dtype: int64
```

Observations 6:

1. Director Noam Murro has directed maximum no. of TV shows which are present on Netflix, followed by Thomas Astruc and Alan Poul

```
df_movies_fil1 = df_movies[df_movies['cast'] != 'Unknown cast']
count_per_cast_Movies = df_movies_fil1['cast'].value_counts()
count_per_cast_Movies.head(10)
```

```
cast
Liam Neeson          161
Alfred Molina        157
```

```
John Krasinski      138
Salma Hayek         130
Frank Langella      128
Anupam Kher         118
John Rhys-Davies    116
Shah Rukh Khan      108
Naseeruddin Shah    106
James Franco        100
Name: count, dtype: int64
```

Observations 7:

1. Actor Liam Neeson is a famous actor which appeared in more that 160 movies present on Netflix.

```
df_TV_fil1 = df_TV[df_TV['cast'] != 'Unknown cast']
count_per_cast_TV = df_TV_fil1['cast'].value_counts()
count_per_cast_TV.head(10)
```

```
cast
David Attenborough    82
Takahiro Sakurai      56
Yuki Kaji             45
Ai Kayano             41
Junichi Suwabe        39
Jun Fukuyama          38
Yuichi Nakamura       38
Daisuke Ono           38
Kate Harbour          37
André Holland         35
Name: count, dtype: int64
```

Observations 8:

1. Actor David Attenborough is a famous actor which appeared in more that 160 TV shows present on Netflix.

```
text = " ".join(df['listed_in'])

wordcloud = WordCloud(width=800, height=400,
background_color='white').generate(text)

# Display the word cloud
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.show()
```

```
df['release_date'] = pd.to_datetime(df['release_year'].astype(str) +
'-01-01')

df['date_diff'] = df['date_added'] - df['release_date']
mode_val = df['date_diff'].mode()
mode_val
```

```
0   547 days
Name: date_diff, dtype: timedelta64[ns]
```

Observations 9:

It takes around 547 days for a movies or TV shows to be added in Netflix after it's release.